LAB # 11: Optical Flow Estimation

<u>Lab Objective:</u>

To implement and analyze two popular optical flow estimation techniques:

- KLT (Kanade-Lucas-Tomasi) Feature Tracker
- Farne-Back Dense Optical Flow Method

Lab Description:

Optical flow refers to the apparent motion of objects, surfaces, or edges in a visual scene when we look at two or more consecutive image frames, such as in a video. This motion can happen because the camera itself is moving, the objects in the scene are moving, or both are moving at the same time. Instead of tracking actual physical movement directly, optical flow analyzes how the intensity patterns (brightness and texture) in the image change from one frame to the next.

In simple terms, optical flow helps us answer the question: *"what is moving, where is it moving, and how fast is it moving?"* It represents this motion using small vectors (arrows) that show the direction and magnitude of movement for different parts of the image.

Optical flow is an important concept in computer vision because it allows machines to understand dynamic scenes. For example, it can be used to detect moving objects, estimate speed, track motion, and even understand human actions. It also plays a key role in applications like video surveillance, autonomous driving, robotics, and medical imaging.

There are generally two types of optical flow methods: sparse optical flow, where motion is calculated only for selected important points in the image, and dense optical flow, where motion is estimated for every pixel. Different algorithms are used depending on the requirement, such as speed, accuracy, and level of detail.

KLT Optical Flow (Kanade-Lucas-Tomasi Method)

The KLT method is a popular technique used for estimating sparse optical flow, which means it tracks motion only for selected important points in an image rather than all pixels. It is widely used because it is fast and works well for real-time applications.

The KLT method is based on three simple assumptions. First is brightness constancy, which means that the brightness of a pixel does not change between consecutive frames. In other words, even if a point moves, its intensity remains the same. Second is the small motion assumption, which means that points move only slightly between frames, making it easier to estimate their movement accurately. Third is spatial coherence, which assumes that neighboring points in a small region move in a similar way, allowing us to analyze motion within a local window instead of the entire image.

The working of KLT can be understood in three main steps. First, feature selection, where the algorithm identifies "good" points to track, usually corners or highly textured regions, using methods like Shi-Tomasi corner detection. These points are chosen because they are easy to track across frames. Second, optical flow estimation, where the movement of each selected point is calculated using the Lucas-Kanade method within a small neighborhood window. This step determines how much and in which direction each point has moved. Third, tracking, where the positions of these feature points are updated in the next frame, and any points that cannot be tracked reliably are removed.

KLT optical flow is commonly used in applications such as object tracking (for example, tracking vehicles or faces in a video) and motion analysis in areas like sports or surveillance systems.

Horn-Schunck Optical Flow Method

The Horn-Schunck method is a classical approach used for estimating dense optical flow, which means it calculates motion for every pixel in the image. Unlike KLT, which focuses only on selected feature points, this method provides a complete motion field, giving a more detailed understanding of how the entire scene is moving.

This method is based on two main ideas. The first is brightness constancy, similar to KLT, which assumes that the intensity of a pixel remains the same as it moves from one frame to the next. Mathematically, this can be written as:

$$I(x, y, t) \approx I(x + u, y + v, t + 1)$$

Here, (u, v)represents the displacement (motion) of the pixel in the horizontal and vertical directions.

The second idea is the smoothness constraint, which assumes that neighboring pixels move in a similar and smooth manner. This helps in avoiding sudden or noisy changes in motion and produces a more consistent flow field across the image.

The Horn-Schunck method works by solving a global optimization problem that tries to satisfy both brightness constancy and smoothness at the same time. As a result, it generates a smooth and continuous map of motion vectors for the entire image.

This method is useful in applications such as motion segmentation, where moving objects are separated from the background, and action recognition, where human movements or activities are analyzed.

Farneback Optical Flow Method

The Farneback method is also a dense optical flow technique, where motion is estimated for every pixel in the image. Unlike Horn-Schunck, it models the image using polynomial expansion, which allows more robust and accurate motion estimation in many practical scenarios.

This method assumes that small regions of the image can be approximated by quadratic polynomials, and motion is estimated by comparing these polynomial representations between consecutive frames. This makes it more efficient and often more accurate in real-world video data.

Farneback optical flow produces a dense flow field similar to Horn-Schunck, but it is typically faster and more stable in practice, especially when dealing with complex motion.

It is widely used in applications such as motion tracking, video stabilization, object detection, and scene understanding.

Lab Tasks:

1- Implement the KLT (Kanade-Lucas-Tomasi) optical flow method to track feature points between consecutive frames of a video. Your implementation should:

- Load or capture a video sequence.
- Detect good feature points using the Shi-Tomasi corner detection method.
- Compute optical flow using the Lucas-Kanade algorithm.
- Visualize the tracked feature points on the video frames.

2- Implement the Farneback optical flow method to estimate dense motion across the entire image. Your implementation should:

- Load two consecutive frames from a video sequence.
- Compute dense optical flow using the Farneback algorithm.
- Convert the flow into magnitude and direction to represent motion.
- Visualize the flow using color representation to show motion direction and magnitude.